

know attribute this to latent semantic indexing, which Google has been using for a while, but recently increased its weighting."

AI And NLP

Coming back to how AI could help, the answer is that AI could help determine one's intent when one feeds in keywords, and it could help in understanding the contents of a page as well. When both these happen, what you have is a smarter search engine. This could happen through implementations of NLP (Natural Language Processing).

We've talked about NLP earlier in *Digit* (June 2003): it's basically a method by which a computer processes something said in a 'natural language' such as English, as opposed to a computer language, and comes up with something intelligent. That is, when you apply NLP to something like "the world is round", the machine would have an internal representation of that fact. That sentence would not remain four, un-understood words, but would mean something to the machine. They would mean that something called 'the world' has a property, and that that property is called 'round'. The system could also, if the knowledge has been fed in, know what round means; it might be able to deduce that the world is therefore in some way similar to a ball, and so on.

In the future, then, if search engines come up with NLP implementations, we'd have a situation in which they would understand what you're saying. When you ask "When was Britney born", they would know that you're talking about a person called Britney; that you need a date as an answer; and that date is connected with that person called Britney; and finally, that the date in question is a birth date. Note that Google seems to be doing something like this already!

On the other side, what about the pages that have the answer to the question? Those pages, when they're crawled or 'spidered' have to have NLP applied to them too, at least in a rudimentary way. If a Web page says "Britney Spears' birthday is the 1st of January", the NLP-enabled spider would be able to deduce, as in the example above, that there is a person called Britney Spears, that that person has a property called a birth date, and that the value of that property is "1st January". The search engine would then serve up the page.

Of course, it's not as simple as it sounds. Conceptually, it is—but NLP is one of the hardest AI problems. However, Tom Mitchell, former president of the AAAI (American Association of Artificial Intelligence), said in November 2003 that in three to five years, we could have something like this.

Mitchell said that we can already develop computer software that can examine a Web page, and find names of people, dates and locations.

"It can't read text and understand it in the level of detail people can, but already, it can read text and can say, 'Oh, this is the name of a person' with about 95 per cent accuracy and, 'Oh, this is a location; this is a date,'" he said.

Researchers have written programs that can

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